

Tanvir Ahmed

B.Sc. in IoT and Robotics Engineering

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🌐 Tanvir Ahmed 🌐 Tanvir Ahmed



Education

B.Sc. Engineering in Internet of Things and Robotics Engineering

2022 – 2026

Institution: *University of Frontier Technology, Bangladesh*

Result: CGPA 3.76/4.00

Higher Secondary Certificate

2020

Institution: *Bangladesh Navy School and College, Chittagong*

Result: GPA 5.00/5.00

Secondary School Certificate

2018

Institution: *BAF Shaheen School and College, Chittagong*

Result: GPA 5.00/5.00

Research Interest

• Machine Learning • Internet of Things and IoT Security • Robotics • Edge AI • Deep Learning

Publications

Journal Publications

1. **Tanvir Ahmed**, Akhterujjaman Siddiquee, Sheikh Sidratul Muntaha Punno, Md. Moshir Rahman, Fahmida Ahmed Antara, Sadia Enam, “IoT-Based Smart Homes: Technologies, Security Risks and Countermeasures,” *International Journal of Information Engineering and Electronic Business (IJIEEB)*, vol. 18, no. 3, June 2026. [IJIEEB](#)

Conference Papers

1. Sheikh Sidratul Muntaha Punno, **Tanvir Ahmed**, Md. Akhterujjaman Siddiquee, and Suman Saha, “Transformer-Based Models for Student Mental Health Detection: A Comparative Study of BERT, RoBERTa and Gemma,” *IEEE 5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE)*, 2026. [IEEE Xplore](#)
2. Sheikh Sidratul Muntaha Punno, **Tanvir Ahmed**, MD. Akhterujjaman Siddiquee, Zohaira Khan Amisha, and Saurav Chandra Das, “AgriMind: An IoT-Driven LLM Framework for Intelligent Precision Agriculture,” *IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, 2026. [IEEE Xplore](#)

Manuscripts Under Review

Journal Papers

1. Sheikh Sidratul Muntaha Punno, **Tanvir Ahmed**, Md. Moshir Rahman, Farzana Akter, and Sadia Enam, “SmartVision: A Multimodal Mobile Assistant for Visually Impaired Users Using Hybrid AI and Real-Time Interaction,” submitted to *Signal, Image and Video Processing* (Springer), 2026.

Bachelor Thesis

Title: A Quantum-Inspired Fidelity-Density Framework with Confidence-Aware Swarm Collaboration for Decentralized IoT Anomaly Detection

- Developed a benign-reference quantum-inspired anomaly detection framework combining fidelity similarity and local density deviation for decentralized IoT intrusion detection.
- Implemented confidence-aware swarm peer collaboration for inference-time alarm verification using confidence-weighted peer consensus and local anomaly evidence.
- Evaluated the framework on CICIoT2023 and X-IIoTID with one-class baselines, repeated-seed experiments, statistical analysis, and an NS-3 network-level validation.

Bachelor Capstone

Title: SmartVision: Multimodal Mobile Assistant for Visually Impaired Users

- Developed a cross-platform mobile assistive application using Flutter to support visually impaired users through real-time navigation, scene description, and obstacle detection.
- Integrated Google Gemini API for online scene understanding and navigation guidance, and TensorFlow Lite for offline on-device object detection and proximity awareness.
- Implemented Text-to-Speech (TTS) and haptic vibration feedback to provide accessible real-time auditory and tactile assistance.

Academic Projects

1. IoT-Driven Smart Farming System

- Developed an ESP32-based smart farming system with temperature, humidity, soil moisture, water-level, and air-quality monitoring.
- Integrated ThingSpeak for real-time sensor visualization and control of a water pump, fan, and artificial lighting.
- Implemented Gemini API-based LLM recommendations with a Flutter mobile and web monitoring interface.

2. Student Mental Health Detection

- Developed a student mental health classification system using BERT, RoBERTa, and Gemma-3-4B-it.
- Performed text preprocessing, tokenization, and multi-class classification on student mental health data.
- Applied LoRA-based fine-tuning to efficiently adapt Gemma-3-4B-it for the classification task.

3. Fire Fighting Rescuers Eye Bot

- Developed a firefighting and rescue robot with autonomous obstacle avoidance, fire detection, and extinguishing capabilities.
- Integrated ESP32-CAM, ultrasonic and flame sensors, L298N motor driver, servo motor, Bluetooth, and water pump.
- Implemented manual and autonomous modes with camera-based monitoring and human detection.

4. Smart Tunnel for Maintenance and Security

- Developed an IoT-based tunnel monitoring system for real-time light, air-quality, and security monitoring.
- Integrated BH1750 and MQ135 sensors with ESP32 and ESP32-CAM for environmental sensing and video surveillance.
- Connected the embedded system with the Blynk platform for remote monitoring and control of tunnel ventilation and related devices.

Technical Skills

Programming Languages	C, C++, Python, Java, Kotlin, Dart
Embedded Systems & IoT	Arduino, ESP8266, ESP32, ESP32-CAM, Raspberry Pi.
AI & Machine Learning	Machine Learning, Deep Learning, Natural Language Processing (NLP), LLM fine tuning, Computer Vision, Transformer Models, BERT, RoBERTa
Platforms & Tools	Blynk IoT, Tinkercad, Google Colab, Android Studio, VS Code, PyCharm, Arduino IDE, Cisco Packet Tracer, VMware Workstation, Visual Studio, Firebase, Apache NetBeans, SQLite, MySQL Workbench, Git, GitHub.
Documentation & Design	Overleaf, draw.io, Canva, Microsoft Word, Excel, and PowerPoint

Languages

English (Professional Working Proficiency), **Bangla** (Native)

Achievements & Certifications

Dean’s List, Level 4 Term 1 & Term 2, University of Frontier Technology, Bangladesh	2026
Dean’s List, Level 3 Term 1 & Term 2, University of Frontier Technology, Bangladesh	2025
Job Ready Certification, Wadhvani Foundation – 21st Century Core Employability Skills Program	2023
Project Showcasing, DUET Techfest 2023, DUET Robotics Club	2023
Junior Scholarship	2015

References

Farzana Akter Assistant Professor Department of Internet of Things and Robotics Engineering University of Frontier Technology, Bangladesh Email: farzana0001@uftb.ac.bd Phone: +8801741582031	Md. Toukir Ahmed Assistant Professor Department of Internet of Things and Robotics Engineering University of Frontier Technology, Bangladesh Email: toukir0001@uftb.ac.bd Phone: +8801738093128
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